Laser Welding Parameter Optimization and Porosity Control

Abstract

This study investigates laser welding of structural steel and establishes evidence-linked processing parameters for quality

Introduction

Laser welding quality depends on heat input, shielding and keyhole stability.

Materials and Methods

The base material was Q355 steel with a thickness of 3 mm.

The laser power was 2.4 kW and travel speed was 12 mm/s.

Table 1 Welding parameters

Group	Laser power	Speed
A	2.0 kW	10 mm/s
B	2.4 kW	12 mm/s
C	2.8 kW	14 mm/s

Results

The highest tensile strength was 612 MPa for parameter group B.

Discussion

Porosity formed because shielding gas was trapped during unstable keyhole collapse; stable keyhole behavior reduced p

Conclusions

A laser power of 2.4 kW produced the best combined mechanical performance in this experiment.

References

[1] Zhang et al. Laser welding process monitoring. 2024.

[2] Wang et al. Keyhole stability and pore formation. 2023.